

CYBERVAULT - COMPLETE FUNCTIONAL REQUIREMENTS

USER ROLES

The system shall support five user roles:

Student

- Register and Login
- Search resources
- View resources online
- Download resources
- Save resources for offline access
- Bookmark resources
- Update profile

Lecturer

- All student privileges
- Upload coursework materials
- Upload research papers
- Upload textbooks
- Edit own uploads
- Academic Researcher
- All student privileges
- Upload scholarly publications
- Upload conference papers
- Upload datasets
- Manage own submissions

Library Staff

- Review uploaded materials
- Approve or reject submissions
- Manage metadata
- Categorize resources
- Archive outdated content

Administrator

- Full system access
- User management
- Department management
- Analytics and reporting
- Backup management
- Repository configuration
- Course Management

USER MODULES

Authentication Module

Registration

Users must select a role:

- Student
- Lecturer
- Academic Researcher

Login

- Email
- Password

Password Recovery

- Email verification
- Password reset functionality

Profile Management Module

Student Profile

Required Fields:

- Full Name
- Student Email
- Gender
- Undergraduate/Postgraduate
- Level
- Department
- Matriculation Number

Lecturer Profile

Required Fields:

- Full Name
- School Email
- Gender
- Department

- Staff ID

Academic Researcher Profile

Required Fields:

- Full Name
- Institutional Email
- Gender
- Institution
- Research Area
- Academic Qualification

DEPARTMENT MODULE

The Faculty of Computing repository shall contain:

Computer Science & Artificial Intelligence

Resources:

- Coursework
- Academic Papers
- Projects
- Datasets

Data Science

Resources:

- Coursework
- Academic Papers
- Research Datasets
- Projects

Software Engineering

Resources:

- Coursework
- Academic Papers
- System Documentation
- Projects

ICT & Cybersecurity

Resources:

- Coursework
- Academic Papers
- Security Reports
- Projects

REPOSITORY MODULE

The repository shall support the storage and management of:

Academic Papers

- Journal Articles
- Conference Papers
- Technical Reports

Research Publications

- Faculty Publications
- Research Papers
- Institutional Publications

Student Research

- Undergraduate Projects
- Postgraduate Projects
- Theses
- Dissertations

Learning Materials

- Lecture Notes
- Textbooks
- Practical Manuals
- Tutorials

Assessments

- Assignments
- Past Questions
- Course Projects

Datasets

- Machine Learning Datasets
- Research Data
- Statistical Data

RESOURCE UPLOAD MODULE

(Web Application Only)

Authorized users shall upload:

Supported Formats

- PDF
- DOCX
- PPTX

Upload Metadata

- Title
- Author(s)
- Department
- Category
- Keywords
- Abstract
- Publication Year
- Resource Type
- File

RESOURCE APPROVAL WORKFLOW

Step 1:

User uploads resource.

Step 2:

Resource status becomes:

Pending Review

Step 3:

Library Staff/Admin reviews submission.

Step 4:

Submission is:

- Approved
- Rejected
- Returned for Revision

Step 5:

User receives notification.

Resource States:

- Draft
- Pending Review
- Approved
- Rejected
- Archived

SEARCH AND DISCOVERY MODULE

Global Search

Search by:

- o Title
- o Author
- o Keywords
- o Department
- o Publication Year

Coursework Search

Search by:

- Course Code
- Course Title
- Level
- Semester

Academic Paper Search

Search by:

- Author
- Title
- Year
- Keywords

Sorting Options

- Newest First
- Most Downloaded
- Most Viewed
- Alphabetical

DOWNLOAD MODULE

Users can:

- Download resources
- View online
- Save for offline access
- View download history

Mobile Features:

- Offline reading
- Download manager

NOTIFICATION MODULE

Notifications shall be sent when:

- Account approved
- Upload approved
- Upload rejected
- New departmental content uploaded
- Password reset requested

Notification Channels:

- In-App Notifications
- Email Notifications

ADMINISTRATIVE DASHBOARD

User Analytics

Display:

- Total Users
- Active Users
- New Registrations

Repository Analytics

Display:

- Total Documents
- Total Downloads
- Most Downloaded Resources
- Most Viewed Resources
- Recent Uploads

Department Analytics

Display:

- Department Contributions
- Department Downloads
- Resource Distribution

SECURITY REQUIREMENTS

Authentication

- Secure Login
- Password Hashing
- Email Verification

Authorization

- Role-Based Access Control (RBAC)

Data Security

- File Encryption
- HTTPS Communication
- Secure Cloud Storage

Backup

- Daily Database Backup
- Weekly Full Backup

Audit Logs

Track:

- Login Attempts
- Upload Activities
- Download Activities
- Administrative Actions

MOBILE APPLICATION FEATURES

- Quick Search
- Offline Access
- Resource Bookmarking
- Push Notifications
- Mobile PDF Viewer
- Download Manager
- Dark Mode
- Low Data Mode

NON-FUNCTIONAL REQUIREMENTS

Performance

- Search response below 3 seconds
- Support 1,000+ concurrent users

Reliability

- Minimum 99% uptime

Scalability

- Support future faculty expansion
- Support university-wide deployment

Usability

- Responsive UI
- Mobile-first design
- User-friendly navigation

DATABASE TABLES

Users

Roles

Departments

Document's

Categories

Downloads

Notifications

Approvals

Audit Logs

Bookmarks

Activity Logs

SUCCESS METRICS

The system will be considered successful when:

- Resources can be retrieved within 3 seconds.
- Mobile and web synchronization functions correctly.
- More than 95% of system tests pass.
- User satisfaction exceeds 85%.
- System uptime remains above 99%.
- Academic materials are accessible from both web and mobile platforms.

FUTURE ENHANCEMENTS

- **AI-Powered Recommendations**
- **OCR Support**
- **Citation Generator (APA, MLA, IEEE)**
- **DOI Integration**
- **Research Collaboration Features**
- **Automated Metadata Extraction**
- **Integration with University Library Systems**
- **Research Impact Analytics**
- **University-Wide Repository Expansion**